



LAHORE GARRISON UNIVERSITY

SOFTWARE VERIFICATION & VALIDATION

QUIZ NO 5&6

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LAHORE GARRISON UNIVERSITY

SECTION A – VALIDATION CHECKLIST ENGINEERING

Q1 (1): Measurable Validation Criteria

Stakeholder expectations → convert into measurable testable criteria:

1. Secure Login

- Password must be encrypted
- Login must block after 5 failed attempts
- Authentication response time ≤ 2 seconds

2. Transaction Accuracy

- Transaction amount must match database entry (100% consistency)
- No duplicate transactions allowed
- Balance update accuracy = 100%

3. Session Timeout

- Session must expire after 10 minutes of inactivity
- User must be logged out automatically
- Re-login required after timeout

4. Mobile Responsiveness

- UI must support screen sizes 320px–1440px
- No horizontal scroll on mobile
- Page load time ≤ 3 seconds on 4G

5. Audit Logging

- All transactions must be logged
- Logs must include timestamp, user ID, action
- Logs must be immutable (no edit/delete)

Q1 (2): Validation Checklist Table

ID	Feature	Validation Criteria	Test Method	Expected Result	Status
1	Secure Login	Encrypt password + 5 failed attempts lock	Login test	Account locked after 5 attempts	Pass/Fail
2	Transaction Accuracy	Balance must update correctly	Transaction simulation	Correct updated balance	Pass/Fail
3	Session Timeout	Logout after 10 min inactivity	Idle test	Auto logout occurs	Pass/Fail
4	Mobile UI	Responsive layout	Device testing	UI fits all screens	Pass/Fail
5	Audit Logging	All actions logged	Log inspection	Logs stored correctly	Pass/Fail

Section B – CI Validation (GitHub Actions)

Q2 (1): CI Validation Explanation

Validation checkpoints in CI pipeline:

- Every code push triggers automated pipeline
- Code is tested using unit/integration tests
- Security checks (like OWASP ZAP or linters) run automatically
- Reports are generated after each run
- Artifacts stored for audit and compliance

Purpose:

- Early bug detection
- Continuous security validation
- Proof of compliance
- Automated documentation